Akash Venkateshwaran

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Data Scientist with 3+ years of experience in machine learning, computational optimization, and data engineering for applied science and engineering. Proven record in developing statistical models, automating data pipelines, and performing rigorous data analyses, supported by successful research projects, internships, and publications.

Technical Skills

Languages: Python, SQL, Bash
Libraries: PyTorch, PyTorch Forecasting, Torchvision, scikit-learn, XGBoost, Catboost, Keras, NumPy, Pandas, SciPy
Tech Stack: Databricks, Apache Airflow, PySpark, Docker, Kedro, MLflow, Git, GitHub Actions, Ray (Data/Train/Tune)

Experience

Machine Learning Resident — August 2025 – August 2026
Alberta Machine Intelligence Institute — *(PySpark · Databricks · MLflow)*

- Built and deployed end-to-end backend pipelines on Databricks, powering an unscheduled shop removal event-series prediction module for the full fleet of 70K+ Pratt & Whitney Canada's engines
- Built PySpark ETL pipelines ingesting 10+ Delta table sources to construct feature tables, parallelizing model training, inference and reporting pipelines using the Kedro framework
- Implemented data profiling, data validation, and data quality checks using Databricks DQX
- Architected a parameterized dynamic model-routing system orchestrating 20+ parallelized models registered in Unity Catalog via MLflow, with a rule-based routing batch inference pipeline serving fleet-wide predictions at scale

Research Engineer — September 2022 – May 2025
University of British Columbia, Vancouver, Canada — *(Python · PyTorch · Ray)*

- **ML Research:**
  - Developed a pioneering conditional CNN achieving 90% SSIM accuracy for far-field acoustic prediction
  - Designed and implemented a data pipeline to preprocess and integrate 3D numerical acoustic propagation and GEBCO bathymetry datasets for model training
  - Developed replay-based continual learning pipeline using stratified memory buffer and reservoir sampling to preserve knowledge across sequential tasks and enhance model generalizability
  - Built end-to-end DeepAR model for multi-variate time-series forecasting of oceanographic data
  - Implemented backtesting framework and systematic hyperparameter optimization using Optuna
  - Performed all distributed model training and orchestration using Ray and MLflow on Sockeye HPC
- **Operations Research:**
  - Modeled a route and speed optimization problem focused on reducing ship noise signature using BIT* and GA
  - Optimized vessel scheduling in the Port of Vancouver to achieve a 70% reduction in acoustic footprint
  - Developed software platform featuring an integrated data-driven forecasting model for proactive voyage planning and vessel traffic management in collaboration with Clear Seas

Data Science Intern — September 2024 – December 2024
IPEX Technologies Inc., Mississauga, Canada — *(GPyTorch · SQL · Hex · scikit-learn)*

- Implemented GPR models within Hex's workspace integrated with Snowflake for polymer property prediction
- Created an interactive knowledge graph dashboard in Hex, mapping chemical relationships and material ingredient dependencies and properties across polymers in the catalog

Publications

- A physics-guided probabilistic surrogate modeling framework for digital twins of underwater radiated noise, *Ocean Engineering*, vol. 358, Jan 2026. I. K. Deo, **A. Venkateshwaran**, and R. K. Jaiman.
- Predicting transmission loss in underwater acoustics using continual learning with range-dependent conditional convolutional neural networks, *Journal of the Acoustical Society of America*, vol. 157, May 2025. I. K. Deo, **A. Venkateshwaran**, and R. K. Jaiman.

Education

University of British Columbia — 2022 – 2025
M.A.Sc. in Engineering, Grades: 90.2% — *Vancouver, Canada*

Vellore Institute of Technology — 2018 – 2022
B.Tech in Engineering, Grades: 96.2% — *Chennai, India*